

Flow Matching · arXiv 2210.02747

Empiricist

11+3 · seat 2

H1-H4 · OT vs VP vs SM · NFE sweep

Sep 23 paper club · VLA Architectures

Spine: OT-CFM objective + path design for action statistics

Lipman et al. · Meta AI / Weizmann · pick a path, regress the VF

Hypotheses H1-H4

Empiricist · falsifiers in notes · P0 toys hours-1 day

H1

OT-CFM $\geq$ quality at lower NFE
than VP-CFM / SM

matched arch + budget

H2

OT reaches target metric in
fewer wall-clock hours

straighter $\rightarrow$ easier regress

H3

Cond CFM $>$ pure regression
on toy cond task

OT still stronger path

H4

Action-chunk OT-CFM fewer
steps than DP-style head

similar success rate

Priority matrix

P0

OT vs VP vs SM on 2D + CIFAR-lite; NFE sweep

P1

Conditional FM toy; CNF likelihood (Hutchinson)

P2

$\sigma_{\min}$; action-space OT-CFM vs DP toy

Defer

ImageNet-64/128 (multi-GPU weeks)

Conditional Flow Matching · the objective card

Thm 2: $\nabla \text{CFM} = \nabla \text{FM}$ · never need the marginal VF

Intractable FM

$$L_{FM} = E_{t, p_t} \|v_t(x) - u_t(x)\|^2$$

needs marginal generating field u_t

hard — mix over all data

Tractable CFM ★

$$L_{CFM} = E_{t, q, x} \|v_t - u_t(x|x_1)\|^2$$

per-sample conditional field

same gradients (Thm 2)

OT-CFM recipe (OpenPI / π_0 action expert)

1. Sample path

$$\begin{aligned} t &\sim U(0,1) \\ \varepsilon &\sim N(0,I), \quad a \sim \text{data} \\ a_t &= (1-t)\varepsilon + t a \end{aligned}$$

2. Target VF

$$\begin{aligned} u &\approx a - \varepsilon \\ &\text{(constant direction)} \\ \sigma_{\min} &\text{softens endpoint} \end{aligned}$$

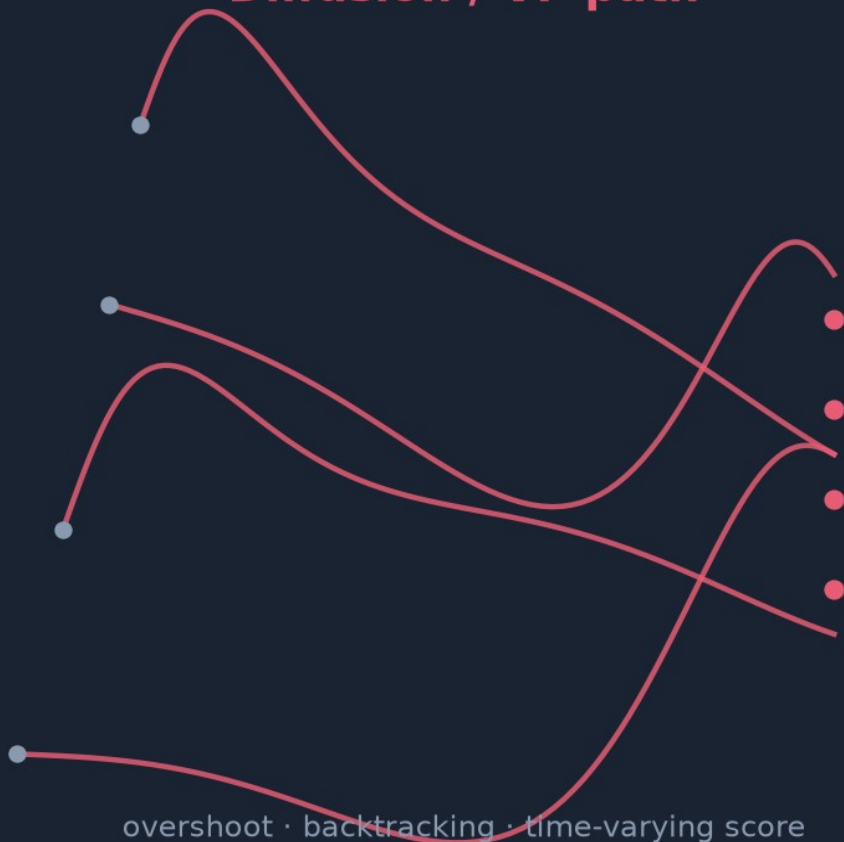
3. Regress

$$\begin{aligned} &\|v_{\theta}(a_t, t | o) - u\|^2 \\ &\text{condition on obs/VLM} \\ &\rightarrow \text{action-chunk expert} \end{aligned}$$

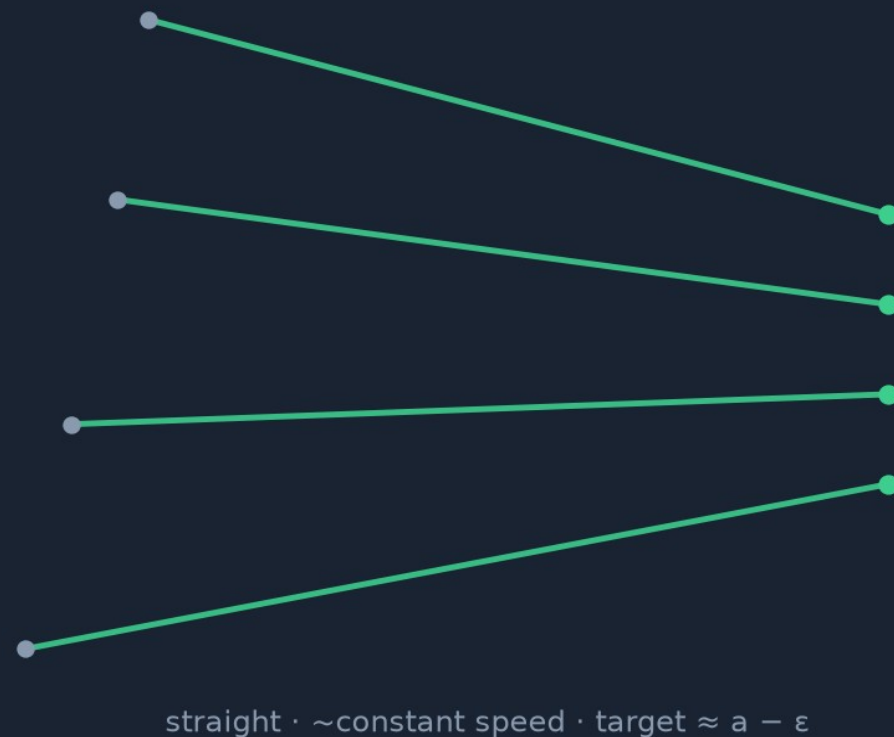
OT straight paths vs diffusion curved paths

Paper Figs 2-3 · constant-direction VF · fewer NFEs for same error

Diffusion / VP path



OT displacement path



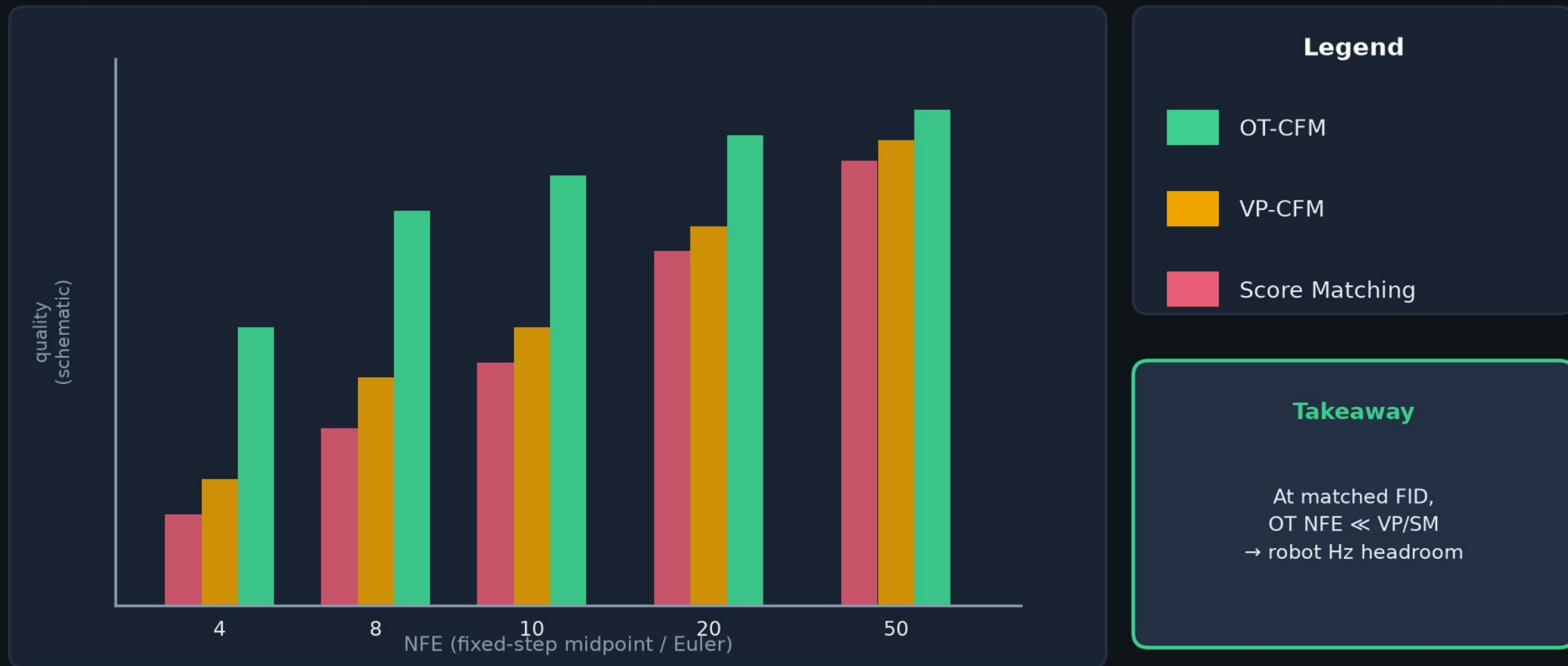


NFE sweep

OT vs VP vs SM

NFE efficiency · OT wins the low-step regime

Paper Fig 7 spirit · FM-OT ~60% NFEs of diffusion for same error · Empiricist H1



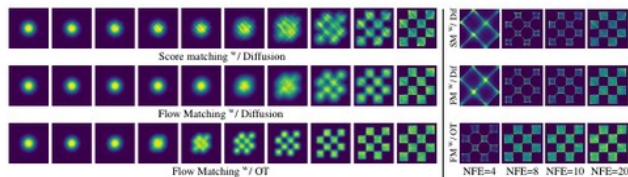


Figure 4: (left) Trajectories of CNFs trained with different objectives on 2D checkerboard data. The OT path introduces the checkerboard pattern much earlier, while FM results in more stable training. (right) FM with OT results in more efficient sampling, solved using the midpoint scheme.

In order to speed up CNF training, some works have developed simulation-free CNF training frameworks by explicitly designing the target probability path and the dynamics. For instance, Rozen et al. (2021) consider a linear interpolation between the prior and the target density but involves integrals that were difficult to estimate in high dimensions, while Ben-Hamu et al. (2022) consider general probability paths similar to this work but suffers from biased gradients in the stochastic minibatch regime. In contrast, the Flow Matching framework allows simulation-free training with unbiased gradients and readily scales to very high dimensions.

Another approach to simulation-free training relies on the construction of a diffusion process to indirectly define the target probability path (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song & Ermon, 2019). Song et al. (2020b) shows that diffusion models are trained using denoising score matching (Vincent, 2011), a conditional objective that provides unbiased gradients with respect to the score matching objective. Conditional Flow Matching draws inspiration from this result, but generalizes to matching vector fields directly. Due to the ease of scalability, diffusion models have received increased attention, producing a variety of improvements such as loss-rescaling (Song et al., 2021), adding classifier guidance along with architectural improvements (Dhariwal & Nichol, 2021), and learning the noise schedule (Nichol & Dhariwal, 2021; Kingma et al., 2021). However, (Nichol & Dhariwal, 2021) and (Kingma et al., 2021) only consider a restricted setting of Gaussian conditional paths defined by simple diffusion processes with a single parameter—in particular, it does not include our conditional OT path. In another line of works, (De Bortoli et al., 2021; Wang et al., 2021; Peluchetti, 2021) proposed finite time diffusion constructions via diffusion bridges theory resolving the approximation error incurred by infinite time denoising constructions. While existing works make use of a connection between diffusion processes and continuous normalizing flows with the same probability path (Maoutsa et al., 2020b; Song et al., 2020b; 2021), our work allows us to generalize beyond the class of probability paths modeled by simple diffusion. With our work, it is possible to completely sidestep the diffusion process construction and reason directly with probability paths, while still retaining efficient training and log-likelihood evaluations. Lastly, concurrently to our work (Liu et al., 2022; Albergo & Vanden-Eijnden, 2022) arrived at similar conditional objectives for simulation-free training of CNFs, while Neklyudov et al. (2023) derived an implicit objective when u_t is assumed to be a gradient field.

6 EXPERIMENTS

We explore the empirical benefits of using Flow Matching on the image datasets of CIFAR-10 (Krizhevsky et al., 2009) and ImageNet at resolutions 32, 64, and 128 (Chrabaszcz et al., 2017; Deng et al., 2009). We also ablate the choice of diffusion path in Flow Matching, particularly between the standard variance preserving diffusion path and the optimal transport path. We discuss how sample generation is improved by directly parameterizing the generating vector field and using the Flow Matching objective. Lastly we show Flow Matching can also be used in the conditional generation setting. Unless otherwise specified, we evaluate likelihood and samples from the model using `dopri5` (Dormand & Prince, 1980) at absolute and relative tolerances of $1e-5$. Generated samples can be found in the Appendix, and all implementation details are in Appendix E.

8 SOCIAL RESPONSIBILITY

Along side its many positive applications, image generation can also be used for harmful proposes. Using content-controlled training sets and image validation/classification can help reduce these uses. Furthermore, the energy demand for training large deep learning models is increasing at a rapid pace (Amodei et al., 2018; Thompson et al., 2020), focusing on methods that are able to train using less gradient updates / image throughput can lead to significant time and energy savings.

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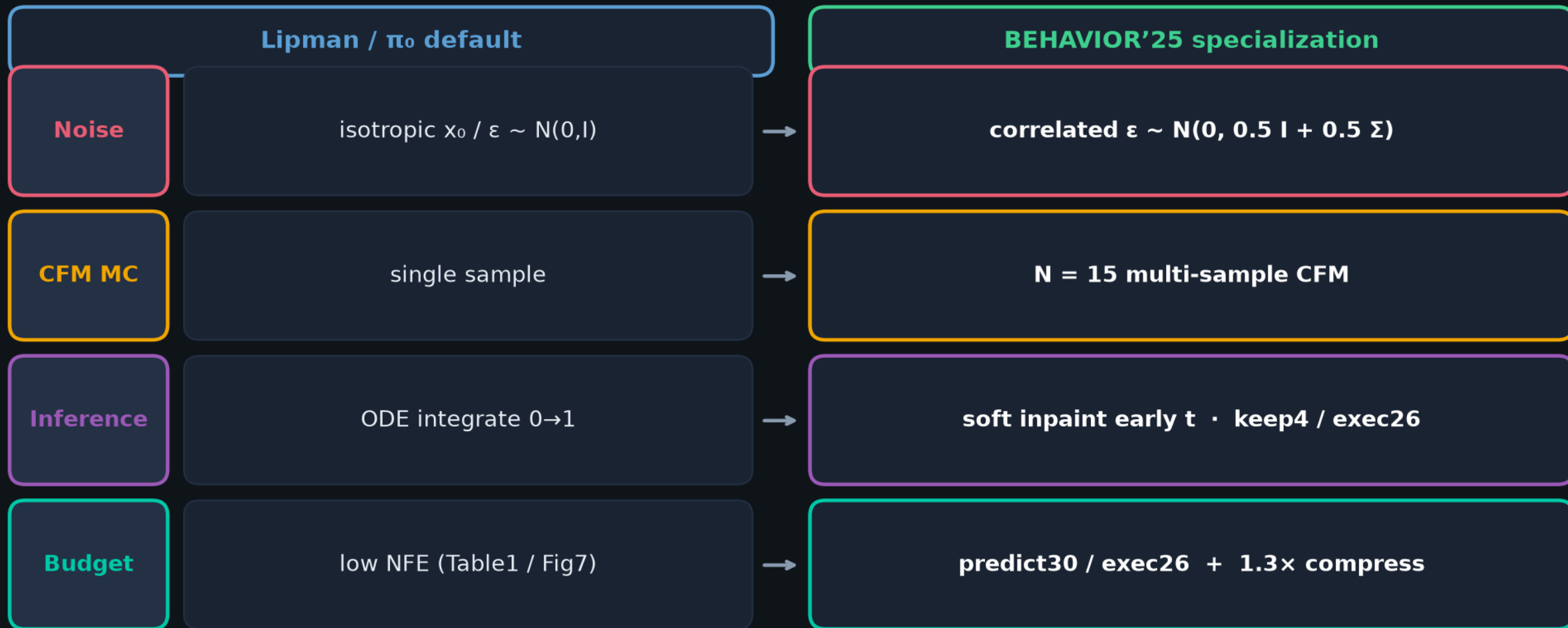
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Toy budget · hours-1 day honesty

Workload	GPU	VRAM	Wall time
2D CFM ablations	0-1 GPU	<2 GB	minutes-1 h
CIFAR-lite OT/VP/SM	1 GPU	6-12 GB	0.5-2 days / run
FID@10k + NFE sweep	1 GPU	4-8 GB	hours
Action OT-CFM vs DP	1 GPU	4-10 GB	hours-1 day
Paper IN-128	32 GPU	large	OUT OF SCOPE

FM → BEHAVIOR map · what changed for robot stats

Lipman'22 foundation → 2025 BEHAVIOR winners on Pi0.5 / OpenPI



Don't overclaim

DP $\neq$ FM

Diffusion Policy = action-space DDPM. Conceptual bridge only. Likely no Lipman cite.

Σ / N=15 $\neq$ Lipman

Correlated noise + multi-sample CFM are BEHAVIOR specializations, not paper defaults.

IN FID $\neq$ robot KPI

ImageNet tables don't transfer to success rate / Hz / contact richness.

Conditional OT $\neq$ marginal OT

Authors admit marginal VF need not be W_2 -optimal — branding can overclaim.

Concurrent literature

Rectified Flow / interpolants overlap — cite them; claim framing+scale+OT emphasis.

Empiricist · close / Q&A

P0 toys first. NFE-quality Pareto under matched nets.

Spine reminder:

FM = pick a path, regress the VF; OT → straighter/faster
→ DP chunks → π_0 OT-CFM head → BEHAVIOR fixes robot stats

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